Hyperspectral Target Detection Using Diffusion Model and Convolutional Gated Linear Unit

Zhaokui Li, *Member, IEEE*, Fan Mo, Xiaobin Zhao, Cuiwei Liu, *Member, IEEE*, Xuewei Gong, Wei Li, *Senior Member, IEEE*, Qian Du, *Fellow, IEEE*, and Bo Yuan

***Abstract*— Deep learning can effectively extract latent information from data to enhance target–background separation in hyperspectral target detection (HTD). However, these models typically require extensive labeled samples, while available target spectra in hyperspectral images (HSIs) are scarce. In addition, existing deep models struggle with target detection in complex backgrounds due to subtle spectral differences. To address these issues, we propose a novel HTD method based on diffusion model and convolutional gated linear unit (HTD-DMCG). First, the diffusion model is integrated with MixUp for data augmentation to generate a diverse and sufficiently large sample set. Next, a transformer architecture using a convolutional gated linear unit (GLU) is designed to effectively capture global dependencies and local feature correlations, leading to more discriminative feature representations. In addition, a new target aggregation and background separation loss is introduced, which emphasizes target sample aggregation while increasing the distance between targets and background samples to enhance separability. The HTD-DMCG method is compared against classical and state-of-the-art HTD methods on four real HSI datasets. Extensive experiments show that it can effectively outperform existing methods in target detection performance. The code is available at https://github.com/Li-ZK/HTD-DMCG**

***Index Terms*— Denoising diffusion probabilistic model (DDPM), hyperspectral target detection (HTD), target intraclass aggregation, transformer encoder.**

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Zhaokui Li, Fan Mo, Cuiwei Liu, and Bo Yuan are with the School of Computer Science, Shenyang Aerospace University, Shenyang 110136, China (e-mail: lzk@sau.edu.cn; mofan@stu.sau.edu.cn; liucuiwei@sau.edu.cn; yuanbo2@stu.sau.edu.cn).

Xiaobin Zhao is with the School of Computer and Communication Engineering, University of Science and Technology Beijing, Beijing 100083, China (e-mail: xiaobinzhao@ustb.edu.cn).

Xuewei Gong is with the CAS Key Laboratory of Forest Ecology and Silviculture, Institute of Applied Ecology, Chinese Academy of Sciences, Shenyang 110016, China (e-mail: gongxw@iae.ac.cn).

Wei Li is with the School of Information and Electronics and the Beijing Key Laboratory of Fractional Signals and Systems, Beijing Institute of Technology, Beijing 100081, China (e-mail: liwei089@ieee.org).

Qian Du is with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS 39762 USA (e-mail: du@ece.msstate.edu).

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I. Introduction

HYPERSPECTRAL remote sensing technology can capture nearly continuous spectral reflectance curves of ground cover, spanning visible, near-infrared, and short-wave infrared bands. Due to the unique physical and chemical properties of each type of ground object, their spectral reflectance curves show significant differences. This spectral fingerprint characteristic allows hyperspectral images (HSIs) to effectively distinguish and identify different targets. Building upon this property, HSIs have facilitated the development of various remote sensing tasks, including classification, super-resolution [1], change detection [2], target detection [3], [4], [5], and anomaly detection [6], [7]. Among these, target detection has emerged as a key research focus in the field of hyperspectral remote sensing [8], [9], [10], owing to its scientific value and practical significance. Compared with high-resolution image target detection that relies on spatial geometric information, hyperspectral target detection (HTD) focuses more on using spectral information to identify targets that are difficult to distinguish through spatial information. This fine-grained analysis allows HTD to maintain high accuracy in complex backgrounds. In recent decades, HTD technology has been widely used in practical applications such as precision agriculture, mineral exploration, urban environment target detection, and military reconnaissance, demonstrating its application value and development potential [11], [12].

The key to HTD lies in developing efficient detectors that can effectively use the spectral differences between various objects to highlight targets while effectively suppressing background interference. The existing HTD methods can be roughly divided into three categories: classical detection methods, machine-learning-based methods, and deep-learning-based methods.

1) Classical Detection Methods: The early HTD detector mainly uses spectral differences between targets and backgrounds, such as spectral angle mapper (SAM) [13] and spectral information divergence (SID) [14] methods. SAM calculates the angle between a testing pixel and target spectra, while SID calculates probability differences between spectra. However, these spectral similarity-based methods are susceptible to noise interference, affecting detection accuracy. Subsequently, detection methods based on signal detection theory were developed, including methods using linear mixing models and statistical distribution assumptions. Classical algorithms in linear mixing models, such as constrained energy minimization (CEM) [15], achieve detection by constructing

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linear filters that maximize target signals while minimizing background signals. In addition, the orthogonal subspace projection (OSP) method [16] projects hyperspectral signals onto subspaces where target–background contrast can be enhanced. Statistical distribution models such as adaptive matched filter (AMF) [17], matched subspace detector (MSD) [18], and adaptive coherent estimator (ACE) [19] have been widely used over the past few decades. Among them, ACE is regarded as one of the most effective detection methods as it treats HTD as a binary hypothesis testing problem through a generalized likelihood ratio test. However, the prior statistical assumptions these methods rely on often do not match reality, resulting in difficulties in meeting actual needs when image scenes are complicated.

2) Machine-Learning-Based Methods: Machine learning techniques are widely used in HTD. Kernel methods, such as kernel-based constrained energy minimization (KCEM) [20] and kernel spectral matched filter (KSMF) [21], project the data into a higher dimensional space to make them linearly separable. Representation-learning-based detectors, such as sparse representation target detectors (SRTDs) [22], sparse representation and binary hypothesis (SRBBH) [23], and combined sparse and collaborative representation (CSCR) [9], represent a pixel as a linear combination of background or target dictionaries and use the representation residual for detection. Some methods, such as spatially adaptive sparse representation for HTD (SASRTD) [24] and weighted Cauchy distance graph (WCDG) and local adaptive collaborative representation [25], also incorporate spatial information. Other machine-learning-based approaches include metric learning [26], multimetric learning [27], multiple instance learning [28], [29], and methods based on low-rank properties and tensor decomposition [30], [31], [32].

3) Deep-Learning-Based Methods: Deep learning methods have rapidly evolved in HTD due to their powerful feature extraction capabilities. Integrating convolutional neural networks into HTD frameworks has become mainstream. For example, Zhu et al. [33] proposed a two-stream convolutional network method (TSCNTD) for HTD. Zhang et al. [34] used pixel pairs to train similarity discrimination models. Some researchers used autoencoders to extract image features or bands for subsequent detection. For instance, Xie et al. [35] proposed a detector combining spectral generative adversarial networks. Li et al. [36] designed hierarchical denoising autoencoders to reduce interference and performed accurate HTD through two-step subspace projection. In addition, spatial features have been incorporated into HTD. Qin et al. [37] introduced visual transformers to learn spectral–spatial features, while Yang et al. [38] emphasized preserving local spatial structures while extracting spatial–spectral information. Rao et al. proposed an HTD method based on a Siamese transformer network [39]. Recently, Liu et al. [40] proposed a weighted discriminative collaborative competitive representation (WDCCR) model for HTD.

Although deep-learning-based HTD methods have made significant breakthroughs in target detection, key challenges remain. Deep learning models require large amounts of annotated data to prevent overfitting, while labeled samples are often very limited in practical HTD applications. In addition, due to subtle spectral differences, the existing deep models struggle to effectively distinguish targets from complex backgrounds, necessitating the development of specialized model architectures to extract more discriminative features.

Regarding the insufficient target sample issue, the key is to construct adequate training samples using single prior target spectra and HSIs. Current solutions mainly fall into generation and synthesis methods. Generation methods use autoencoder [34] or generative adversarial network [41] to create samples, but these samples are highly similar to prior spectra and cannot simulate subtle spectral differences between pixels. Synthesis methods generate target samples by mixing prior spectra with background samples [33], but the k-means-based random initialization leads to unstable results. Recently, diffusion models [42], [43] have gained attention in image generation and have been applied to hyperspectral image classification [44], super-resolution [45], and anomaly detection [46] in remote sensing. Research shows that diffusion models generate better samples, but in HTD, limited prior target spectra make it ineffective. To ensure sample diversity, we combine diffusion models with the MixUp technique [47], [48], [49], generating enhanced images through weighted combinations of prior and generated targets. This approach has proven effective in computer vision. Through MixUp, we blend prior target spectra with diffusion model-generated target spectra, successfully simulating spectral variations while maximizing retention of prior target spectral information.

For discriminative feature extraction, current deep-learning-based HTD mainly uses CNN to extract spatial and local features [33], [34], but CNN struggles to capture long-range dependencies of spectral features. While transformer can compensate for this with its global dependency modeling capability [37], [39], it often neglects local details, reducing feature discrimination. Previous research has attempted to combine global and local features, such as Jiao et al. [50] proposed group spectral embedding and transformer combination method. However, this method only focuses on neighboring spectral information during spectral embedding and does not perform any filtering on the obtained token-level information, which can lead to issues such as information redundancy. This article proposes a transformer architecture based on convolutional gated linear unit (GLU). With 1-D gated convolution units after multihead attention, it enables each token to perform channel attention based on neighboring features, thus maintaining global long-range dependency modeling while enhancing local feature extraction capability.

Achieving a complete separation between target and background samples remains a challenging problem due to their subtle spectral differences. For better differentiation, it is necessary for target samples to cluster around the prior spectral signature. Therefore, a novel target aggregation and background separation (TABS) loss function is designed. This loss function not only pushes target and background samples apart but also emphasizes the aggregation of target samples, ensuring they cluster around the prior target signature for enhanced target–background separation.

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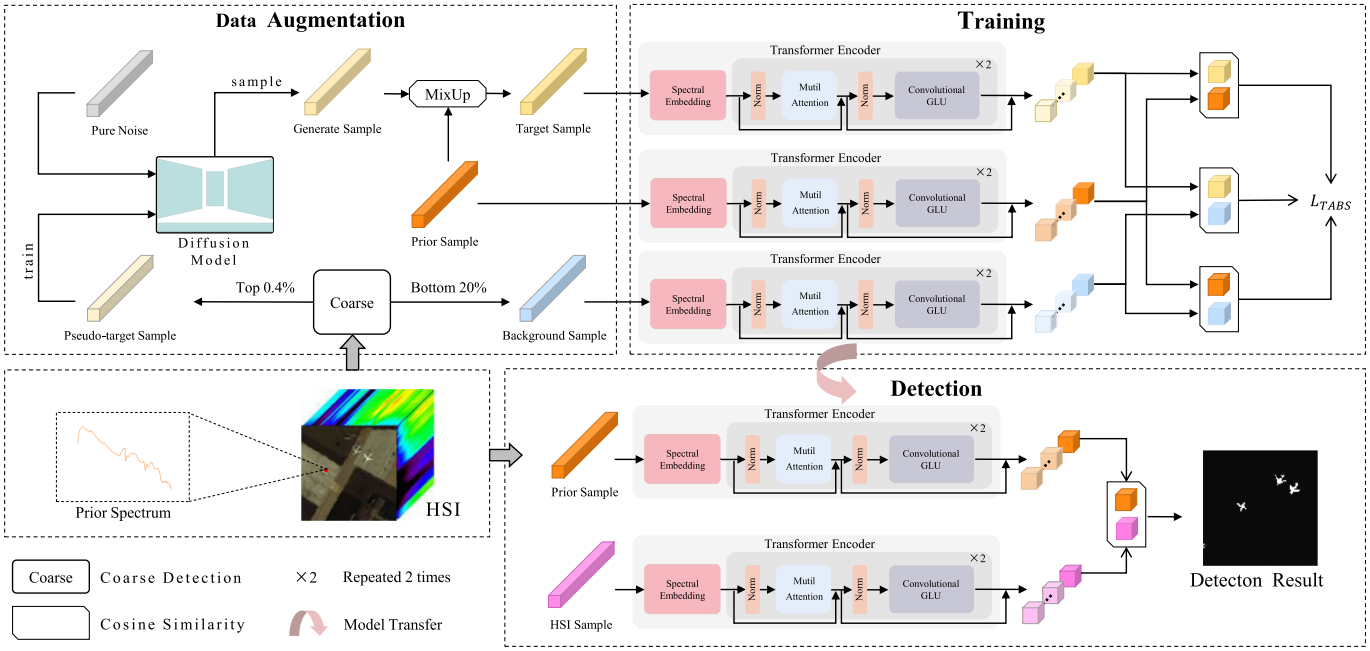


Fig. 1. Flowchart of the proposed HTD-DMCG, which consists of three main parts: data augmentation, training, and detection, is as follows. During the data augmentation phase, a balanced dataset of positive and negative samples is constructed. These augmented samples are then used to train the spectral feature encoder. Finally, the trained encoder is used for detection.

Based on the above analysis, this article proposes a novel HTD method based on diffusion model and convolutional GLU (HTD-DMCG). First, diffusion model and MixUp technique are used for target sample augmentation. Then, a transformer network architecture based on convolutional GLU is designed, implementing channel attention within tokens by introducing 1-Dgated convolution units. In addition, the TABS loss function is designed to separate target samples from background samples while emphasizing the clustering among target samples. The main contributions of the proposed method are summarized as follows.

1) A data augmentation method combining diffusion model and MixUp technique is proposed, which uses diffusion model to supplement insufficient labeled samples and uses MixUp to generate diverse samples containing prior target information, thereby improving detection performance.
2) A transformer architecture based on convolutional GLU is constructed, using convolutional GLU to implement token-level local channel attention mechanism, which can simultaneously capture global dependencies and local feature correlations to extract more discriminative feature representations.
3) A proposed TABS loss function emphasizes the aggregation among target samples while pushing away the distance between targets and background, further enhancing the separability between targets and background.

The remainder of this article is organized as follows. Section II introduces the proposed HTD-DMCG. In Section III, the experimental results and corresponding analysis are presented. Finally, Section IV concludes this article.

II. Proposed Method

In this article, the HSI and the prior target spectrum are defined as $\mathcal{X} \in \mathbb{R}^{h\times w\times l}$ and $x_{\text{prior}} \in \mathbb{R}^{l\times 1}$, where h and w represent the height and width of the HSI (with the total number of pixels in the image given by $n = h \times w$), respectively, and l denotes the number of spectral bands in the HSI.

Fig. 1 illustrates the overall flowchart of the proposed method. First, the prior sample x_{prior} and the HSI $\mathcal{X}$ are fed into a coarse detector to obtain an initial, less reliable detection result, from which pseudotarget samples $x_{\text{pseudotarget}}$ and background samples $x_{\text{background}}$ are selected. The selected pseudotarget sample $x_{\text{pseudotarget}}$ is then used as a training sample for a denoising diffusion probabilistic model (DDPM). The diffusion model is trained on these samples, and generated samples x_{generate} are obtained via the sampling process. Without changing the number of samples, the generated sample x_{generate} is mixed with the prior sample x_{prior} to obtain the target sample x_{target}. The target sample x_{target}, prior sample x_{prior}, and background sample $x_{\text{background}}$ are then used as the positive sample, anchor, and negative sample, respectively, in a triplet loss framework, and are fed into a weight-shared, convolutional GLU-based three-channel transformer encoder for training. The convolutional GLU unit allows the transformer encoder to learn more important features while filtering out irrelevant information. The three-channel feature extractor is constrained by a hybrid loss, enabling the convolutional GLU transformer encoder to learn the differences between target and background samples in the current dataset. This encourages target samples to be closer in feature space while increasing the distance between target and background samples. During the detection phase, the prior sample x_{prior} and the HSI $\mathcal{X}$ are input into the trained convolutional GLU transformer encoder.

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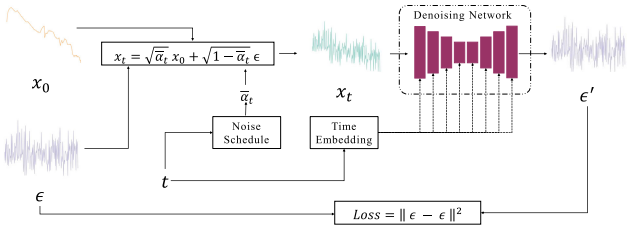


Fig. 2. Training process of DDPM.

The similarity scores between each pixel in the HSI and the prior sample x_{prior} are computed, which are then used as the final detection results.

A. Data Augmentation

1) Coarse Detection: The prior sample x_{prior} and the HSI $\mathcal{X}$ are input into the coarse detector (CEM method) to obtain preliminary, less reliable detection results. These coarse detection results are then used as a reference for selecting the training samples for target and background. Specifically, the top 0.4% of pixels with the highest detection scores are selected as pseudotarget samples $x_{\text{pseudotarget}} \in \mathbb{R}^{l\times s}$, where $s = n \times 0.4\%$, and $s \ll n$. The bottom 20% of pixels with the lowest detection scores are selected as background samples $x_{\text{background}} \in \mathbb{R}^{l\times m}$, where $m = n \times 20\%$.

2) Sampling From the Diffusion Model: As a generative model, the DDPM learns the distribution of the data and progressively transforms random noise into new samples that are aligned with the distribution of the training data. The training process of the DDPM corresponds to one step in the forward process of the diffusion model, while the sampling process of the DDPM is equivalent to the reverse process of the diffusion model.

Due to the use of the U-Net architecture as the denoising network in the DDPM, training samples need to undergo four downsampling and three upsampling steps. Therefore, the pseudotarget samples $x_{\text{pseudotarget}}$ must be processed to obtain the training samples for the DDPM, denoted as $x_{\text{diffusion}} \in \mathbb{R}^{k\times m}$, where k must satisfy $k \mod 8 = 0$. Fig. 2 illustrates the training process of the DDPM. The training sample $x_{\text{diffusion}}$ is then fed into the DDPM, which generates a corresponding number of steps based on the total number of training samples, representing the step at which each training sample is located. Using the training sample and its corresponding step, a sample with added Gaussian noise is generated. This noisy sample at the given step is passed through the denoising network, which predicts the sample from the previous step. Since the forward process of the diffusion model is fully computable, the denoising network can be constrained by minimizing the Euclidean distance between the predicted and actual samples. The sampling process of the DDPM involves generating a Gaussian noise sample with the same dimensions as the training sample. This Gaussian noise sample is then passed through the denoising network multiple times, gradually acquiring the features of the training sample. Eventually, this results in the generation of samples $x_{\text{generate}} \in \mathbb{R}^{k\times m}$ that match the distribution and features of the training data. After simple processing, the generated sample $x_{\text{generate}} \in \mathbb{R}^{l\times m}$ is obtained. It is worth noting that the number of generated samples x_{generate} is consistent with the number of background samples $x_{\text{background}}$ previously selected.

Specifically, to begin with, it is necessary to obtain a DDPM that learns the distribution of training samples $x_{\text{diffusion}}$. The total number of steps for the DDPM is set to T, where it takes T steps for a sample to evolve from the original HSI data to pure Gaussian noise. The set of all the steps is represented as $\{t_0, t_1, t_2, \ldots, t_i\},\ i \in [0, T]$. The noisy samples at different steps are represented by $\{x_0, x_1, x_2, \ldots, x_i\},\ x_i \in \mathbb{R}^{k\times 1},\ i \in [0, 1000]$, where a smaller i indicates less noise has been added. During the training of DDPM, the training sample $x_{\text{diffusion}}$ is represented as x_0, while the pure noise sample is denoted as x_T. Since the forward process of the diffusion model is a Markov process and uses the reparameterization trick, the noisy sample x_t at step t can be obtained from the original sample x_0 and the current step t using the following equation:

$$\begin{aligned} x_t &= \sqrt{\bar{\alpha}_t} \times x_0 + \sqrt{1-\bar{\alpha}_t} \times \epsilon \\ a_t &= 1-\beta_t\ ,\ \ \bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i \end{aligned} \tag{1}$$

where ϵ represents the noise sampled randomly from a Gaussian distribution, i.e., $\epsilon \sim \mathcal{N}(0, I)$, and β_t denotes the variance of the Gaussian noise added at step t which increases as t becomes larger. The term α_t denotes the proportion of the original signal retained at step t. Consequently, $\bar{\alpha}_t$ represents the cumulative product of signal retention ratios from step 1 to step t, indicating the total proportion of the original data preserved up to the current step. A smaller $\bar{\alpha}_t$ implies a higher level of corruption by noise, while a value closer to 1 indicates that the sample remains largely similar to the original input.

The denoising network predicts the noise ϵ' added at the current step. The loss function $L_{\text{diffusion}}$ of the denoising network is the mean-squared error between the true noise ϵ and the predicted noise ϵ', i.e.,

$$L_{\text{diffusion}} = \|\epsilon - \epsilon'\|^2. \tag{2}$$

A sample randomly simulated from the Gaussian noise distribution is denoted as the pure noise sample x_T. It is then fed into the denoising network, which has learned the data distribution of the training samples, to obtain the noisy sample x'_{T-1}. This process is repeated for T steps, resulting in a new sample x'_0, which follows the data distribution of the training samples $x_{\text{diffusion}}$ and has never been encountered before. The generated sample x_{generate} is obtained through this process.

3) Data Mixing: To bring the target samples closer to the prior spectrum in feature space, a data mixing method is used to generate new target samples x_{target}, while maintaining the same number of samples

$$x_{\text{target}} = \gamma \times x_{\text{prior}} + (1-\gamma) \times x_{\text{generate}} \tag{3}$$

where $\gamma \in (0, 1)$ is a random variable following a Beta distribution $\gamma \sim \beta(\omega, \omega)$, where $\omega \in (0, +\infty)$ is a hyperparameter that controls the shape of Beta distribution.

At the final stage of data processing, the generated sample x_{generate}, prior sample x_{prior}, and background sample $x_{\text{background}}$ are used to for a triplet of samples. Specifically, the generated

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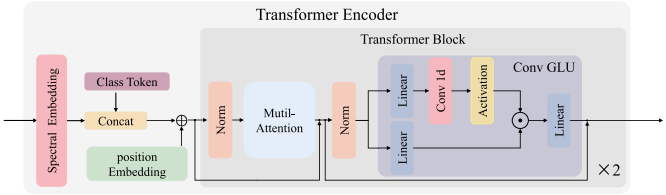


Fig. 3. Structure of the convolutional GLU transformer encoder.

sample x_{generate} is treated as the positive sample, the prior sample x_{prior} as the anchor, and the background sample $x_{\text{background}}$ as the negative sample.

B. Convolutional GLU Transformer Encoder

The transformer architecture, which gained popularity in natural language processing, has been widely applied in computer vision and hyperspectral classification. Inspired by [51], the goal is to obtain more refined feature representations by allowing each token in the convolutional GLU to have a unique gating signal based on its local features. Moreover, the GLU has demonstrated superior performance over multilayer perceptron (MLP) in several tasks. Therefore, a transformer encoder with a gating mechanism is used as the feature extractor in the proposed method. The aforementioned sample triplets are input into a three-channel transformer encoder, where the weights are shared across the three channels. Specifically, for each branch channel, the hyperspectral samples are first partitioned into patches, and a learnable class embedding is added at the beginning of each hyperspectral patch. Then, positional encodings are added to the embedded tokens. The processed tokens are fed into the transformer block, and the learnable class embeddings in the output features are used as the sample features of the input hyperspectral samples. Fig. 3 illustrates the flowchart of the convolutional GLU transformer encoder, and a detailed description is provided below.

Specifically, given a hyperspectral sample $x = x^1, x^2, \ldots, x^l \in \mathbb{R}^{l\times 1}$, the first step is to perform spectral embedding, where nine adjacent spectral bands are grouped as a spectral and undergo dimensionality expansion. After spectral embedding, the hyperspectral sample x is transformed into tokens $y = y^1, y^2, \ldots, y^d \in \mathbb{R}^{d\times 128}$, where d is the number of tokens after the binding of the hyperspectral sample. Next, a learnable class embedding y^{cls} is added, followed by the addition of positional encoding, resulting in the tokens to be input into the transformer block: $y = y^{\text{cls}}, y^1, y^2, \ldots, y^d \in \mathbb{R}^{(d+1)\times 128}$. After passing through multihead attention, the resulting tokens are $z = z^{\text{cls}}, z^1, z^2, \ldots, z^d \in \mathbb{R}^{(d+1)\times 128}$, where

$$z(y) = \text{Attention}(y). \tag{4}$$

Fig. 3 illustrates the detailed structure of the convolutional GLU. To maintain the same depth as the MLP, the input token z is split into two equally sized components, z^1 and z^2, after passing through the first fully connected layer, and these components are fed into separate branches. After passing z^1 through a 1-D convolution and activation function, it is elementwisely multiplied (Hadamard product) with the data z^2 from the other branch. The resulting product is then passed through another fully connected layer to obtain the feature output $\text{Out}(z) \in R^{(d+1)\times 128}$

$$\begin{aligned}\text{Out}(z) = W_2 \cdot (\text{Gelu}(V \cdot (W_1 \cdot z + b_1) + b_3) \\ \odot (W_1 \cdot z + b_1)) + b_2\end{aligned} \tag{5}$$

where W_1 and b_1 represent the weights and biases of the first fully connected layer, respectively, V and b_3 are the weights and biases of the 1-D convolution, respectively, and W_2 and b_2 are the weights and biases of the second fully connected layer, respectively. The function $\text{Gelu}(\cdot)$ is the activation function. The learnable class embedding in $\text{Out}(z)$ is used as the output feature for the given hyperspectral sample.

C. TABS Loss Function

The sample triplet is used to extract features from HSI pixels. Therefore, a novel TABS loss function is proposed to better distinguish between target and background. The features extracted by the transformer encoder for the positive sample (x_{generate}), anchor sample (x_{prior}), and negative sample ($x_{\text{background}}$) are represented as $f(x_p)$, $f(x_a)$, and $f(x_n)$, respectively.

Typically, triplet networks use a fixed distance to ensure a margin between positive and negative samples in the feature space. Jiao et al. [50] designed the ISIA loss, which considers the distance between positive and negative samples, ensuring that negative samples are pushed away from the anchor point while also distancing themselves from positive samples. However, the label information used in this loss may cause the strong and reliable information carried by the prior sample to be overshadowed in the feature space. Given the complexity of background sample categories in HTD tasks, it is crucial to ensure that positive samples are densely distributed around the prior sample in the feature space. Inspired by the improvements to triplet loss discussed in [52], we propose the TABS loss function, specifically designed for HTD tasks. This loss function not only separates target and background but also places emphasis on the distance between the anchor and positive samples, ensuring that positive samples are more tightly clustered around the anchor point.

Cosine similarity $c(\cdot)$ is used to measure the distance between two samples in the feature space. The closer the cosine similarity between two samples is to 1, the closer they are in the feature space; conversely, the farther apart they are. Based on this cosine similarity formula, for the ith triplet, the three samples are paired and compared, resulting in three similarity scores

$$\begin{aligned} c_i^+ &= \max\left(c\left(f\left(x_i^p\right), f\left(x_i^a\right)\right), 0\right) \\ c_i^- &= \max\left(c\left(f\left(x_i^n\right), f\left(x_i^a\right)\right), 0\right) \\ c_i^{-'} &= \max\left(c\left(f\left(x_i^n\right), f\left(x_i^p\right)\right), 0\right) \end{aligned} \tag{6}$$

where c_i^+ denotes the distance between the positive sample and the anchor, c_i^- represents the distance between the negative sample and the anchor, and $c_i^{-'}$ indicates the distance between the negative and positive samples.

Ideally, the loss function should not only focus on the distance between negative samples and the anchor and between negative and positive samples but also aim to reduce the

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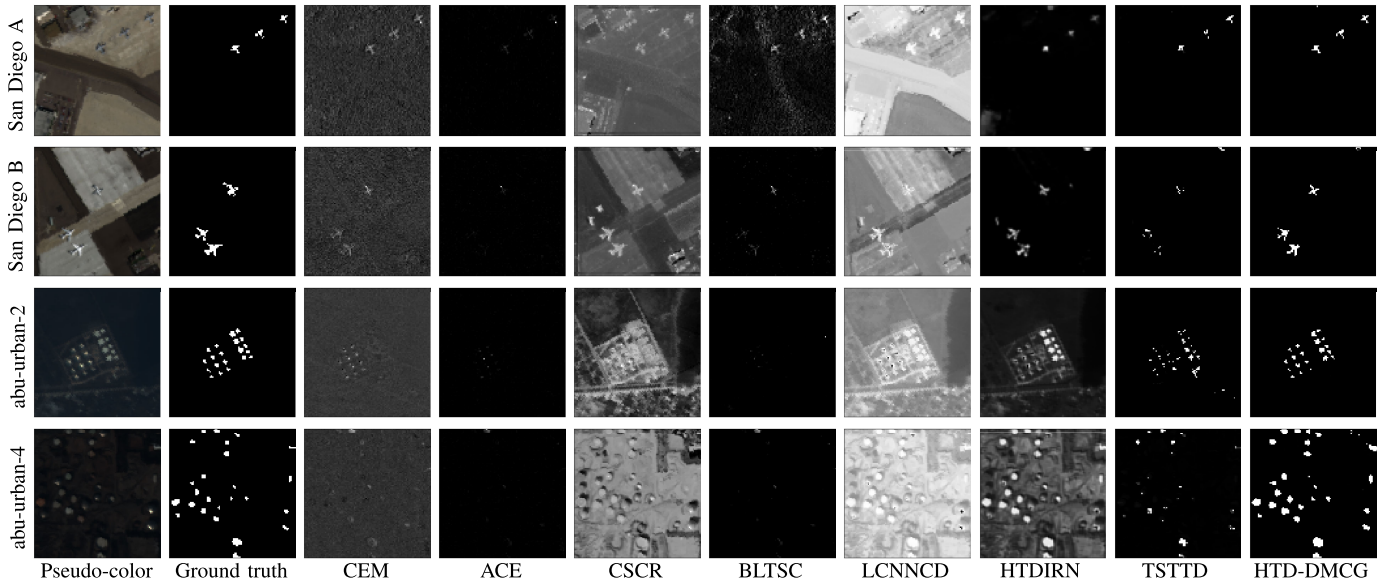


Fig. 4. Visualized results of the competing methods.

distance between the anchor and positive samples. Therefore, the TABS loss function can be divided into three subloss functions, comprising two key terms

$$L_{\mathrm{TABS}} = \underbrace{L_{\mathrm{Triplet}} + \varepsilon \times L_{\mathrm{BCE}}}_{\text{Separation term}} + \underbrace{\beta \times L_{\mathrm{TA}}}_{\text{Aggregation term}} \tag{7}$$

where the triplet loss function L_{Triplet} and binary cross-entropy loss function L_{BCE} serve as the background separation components, ensuring that target and background are effectively separated. On the other hand, the target intraclass loss L_{TA} functions as the target aggregation component, encouraging the target samples to aggregate around the anchor. The hyperparameters ε and β are used to adjust the relative importance of L_{BCE} and the L_{TA} with respect to other components of the loss function. The formula for the triplet loss L_{Triplet} is as follows:

$$L_{\mathrm{Triplet}} = \frac{1}{b}\sum_{i=1}^{b} \max\left(0,\ \mathrm{margin}_1 + \max\left(c_i^-, c_i^{-'}\right) - c_i^+\right) \tag{8}$$

where b represents the number of triplets in a batch and margin_1 is the minimum similarity margin between positive and negative samples. The term $\max(c_i^-, c_i^{-'})$ ensures that the sample with higher similarity to the negative sample in the triplet is used for network learning. This loss function can effectively distinguish between target and background.

The objective of the binary cross-entropy loss L_{BCE} is to position all the target samples as close as possible to 1 in the feature space while positioning all the background samples as close as possible to 0. This process helps achieve target–background separation. Specifically, the labels for positive and negative samples are set to 1 and 0, respectively. In other words, for each triplet, the label for the distance c_i^+ between the positive sample and the anchor is $y_i^+ = 1$, and the label for the distance c_i^- between the negative sample and the anchor is $y_i^- = 0$. The binary cross-entropy loss L_{BCE} is defined as

$$L_{\mathrm{BCE}} = -\frac{1}{2b}\sum_{i=1}^{b} y_i^+ \cdot \log\left[\sigma\left(c_i^+\right)\right] + \left(1 - y_i^-\right) \cdot \log\left[\sigma\left(1 - c_i^-\right)\right] \tag{9}$$

where $\sigma(\cdot)$ represents the sigmoid function.

Due to the inability to directly assess the reliability of the current partitioning of positive and negative samples, and the complex and diverse nature of background sample classes, it is necessary to preserve more reliable information. Thus, the target intraclass loss L_{TA} is designed to measure the distance between the target sample and the anchor

$$L_{\mathrm{TA}} = \max\left(0,\ \mathrm{margin}_2 - c_i^+\right) \tag{10}$$

where margin_2 is the minimum similarity threshold between the anchor and the positive sample, which aims to maintain the compactness of target samples.

III. Experimental Results and Analysis

A. Datasets

The effectiveness of the proposed method is validated using four real datasets: San Diego A, San Diego B, abu-urban-2, and abu-urban-4. Fig. 4 presents the pseudocolor images and ground-truth images of these datasets. In the context of HTD, the spectral characteristics of the target are assumed to be known. In this study, following the approach of [53], the prior spectra used are the pixels closest to the average spectral curve in the current HSI.

1) The San Diego A and San Diego B HSIs were captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) over the San Diego Airport in California, USA. Each subimage of these hyperspectral datasets contains 100 × 100 pixels, with a spatial resolution of 3.5 m. The San Diego B dataset, after removing low signal-to-noise ratio (SNR) and water-absorbing

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bands (1–6, 33–35, 97, 107–113, 153–166, and 221–224), retains 189 spectral bands within the 400–2500-nm wavelength range, with a spectral resolution of 10 nm. The targets in both San Diego A and San Diego B are aircraft, consisting of 58 and 134 pixels, respectively.

2) The abu-urban-2 and abu-urban-4 HSIs are two real-world datasets from the airport–beach–urban collection. The airport–beach–urban dataset was compiled by Professor Kang Xudong from Hunan University, based on data collected from the AVIRIS sensor and publicly available online datasets. The abu-urban-2 and abu-urban-4 images were captured over areas in Texas and Los Angeles, California. These images, each with a 100×100 pixel resolution, have spatial resolutions of 17.2 and 7.1 m, respectively. After removing some SNR and water-absorbing bands, the abu-urban-2 and abu-urban-4 datasets retain 207 and 205 spectral bands, respectively, with a spectral resolution of 10 nm. The targets to be detected in both abu-urban-2 and abu-urban-4 are man-made buildings, with 155 and 272 pixels in each image, respectively.

B. Evaluation Metrics

The entire HSI, along with the hyperspectral data of a single prior target pixel, is input into the HTD algorithm, producing a grayscale image with the same spatial dimensions as the HSI. The value of each pixel in this grayscale image represents the similarity between pixels and the target signature. Generally, the higher the pixel value in the grayscale image, the higher the likelihood that the corresponding pixel belongs to the target class; conversely, lower values indicate that the pixel belongs to the background.

The receiver operating characteristic (ROC) curve is widely used as an evaluation metric HTD. By setting different thresholds τ, the x-axis of the curve represents the probability of background samples being incorrectly detected as target samples, which is known as the false alarm rate P_{F}. The y-axis represents the probability that target samples are correctly identified as target samples, which is referred to as the true positive rate P_{D}. Given a threshold τ, the false alarm rate P_{F} and true positive rate P_{D} of the traditional ROC curve can be calculated using the following formulas:

$$P_{\mathrm{D}} = \frac{N_d}{N_t}, \quad P_{\mathrm{F}} = \frac{N_f}{N_b} \tag{11}$$

where N_d represents the number of target samples correctly identified as targets, N_t denotes the total number of true target samples, N_f indicates the number of background samples incorrectly identified as targets, and N_b refers to the total number of true background samples. Generally, the closer the ROC curve is to the top-left corner, the better the detection performance.

To mitigate the risk of reduced detection performance due to insufficient experience in setting thresholds, a 3-D ROC curve is introduced as a detection metric [54]. This involves incorporating threshold data to display the ROC curve under different thresholds. Furthermore, the 3-D ROC curve leads to the derivation of two curves: $\mathrm{ROC}_{(\tau,\ \mathrm{P_D})}$ and $\mathrm{ROC}_{(\tau,\ \mathrm{P_F})}$. The $\mathrm{ROC}_{(\tau,\ \mathrm{P_D})}$ curve reflects the detection performance at various threshold levels, while the $\mathrm{ROC}_{(\tau,\ \mathrm{P_F})}$ curve indicates the suppression of background at different thresholds. Generally, the 3-D ROC curve's performance improves as it shifts along the coordinate axes. The closer the $\mathrm{ROC}_{(\tau,\ \mathrm{P_D})}$ curve is to the top-right corner, the better the detection performance. Similarly, the closer the $\mathrm{ROC}_{(\tau,\ \mathrm{P_F})}$ curve is to the bottom-left corner, the better the background suppression effect.

To quantitatively evaluate the performance of HTD algorithms, the areas under the three aforementioned 2-D ROC curves are calculated, denoted as $\mathrm{AUC}_{(\mathrm{P_F},\ \mathrm{P_D})}$, $\mathrm{AUC}_{(\tau,\ \mathrm{P_D})}$, and$\mathrm{AUC}_{(\tau,\ \mathrm{P_F})}$. Given the characteristics of the ROC curve, the AUC values for HTD algorithms range from 0 to 1. Generally, the closer the values of $\mathrm{AUC}_{(\mathrm{P_F},\ \mathrm{P_D})}$ and $\mathrm{AUC}_{(\tau,\ \mathrm{P_D})}$ to 1, and the closer the value of $\mathrm{AUC}_{(\tau,\ \mathrm{P_F})}$ to 0, the better the detection performance.

In recent years, most HTD algorithms have reached values near 0.99, making it difficult to distinguish the performance of different algorithms. To address this issue, $\mathrm{AUC_{OA}}$ and $\mathrm{AUC_{SNPR}}$ are proposed [54], as

$$\mathrm{AUC_{OA}} = \mathrm{AUC}_{(\mathrm{P_F},\ \mathrm{P_D})} + \mathrm{AUC}_{(\tau,\ \mathrm{P_D})} - \mathrm{AUC}_{(\tau,\ \mathrm{P_F})} \tag{12}$$

$$\mathrm{AUC_{SNPR}} = \frac{\mathrm{AUC}_{(\tau,\ \mathrm{P_D})}}{\mathrm{AUC}_{(\tau,\ \mathrm{P_F})}}. \tag{13}$$

Here, $\mathrm{AUC_{OA}}$ ranges from 0 to 2, with values closer to 2 indicating better detection performance. $\mathrm{AUC_{SNPR}}$ takes values in the range of $(0, +\infty)$, with a larger value indicating superior detection performance.

Box plots can assess the degree of separation between target and background. A red box represents the target class, while a blue box represents the background class. In the same scene, a larger gap between the red and blue boxes and a shorter blue box indicate a higher degree of target–background separation.

C. Experimental Setups

The denoising network uses a U-Net architecture, and modifications are made to the training samples of DDPM. Specifically, without altering the features, several copies of the last spectral band are appended to the end of each training sample, with the number of copies depending on the number of bands in the dataset. Prior to training the feature extraction network, the denoising network must be trained to generate sufficient training samples, which requires some time. For the feature extraction network used, the spectral embedding maps each divided spectral band to 128 dimensions with two transformer blocks. During network training, the Adam optimizer is used to minimize the loss. Given the task setup for HTD, there is no need to divide the HSI into training and testing sets. The learning rates for San Diego A, San Diego B, abu-urban-2, and abu-urban-4 are set to $1e^{-3}$, $1e^{-4}$, $1e^{-4}$, and $1e^{-4}$, respectively. Considering the impact of the parameters margin_1, margin_2, β, and ε on detection performance, these parameters will be analyzed and determined in Section III-E. It is worth noting that in the abu-urban-4 dataset, selecting too many samples would introduce a large number of noisy samples into the diffusion model. Therefore, in this scenario, only the top 0.1% of samples based on coarse detection scores are used for training the denoising network. Before training

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the feature extractor, the denoising model must be trained and training samples must be generated, which requires additional time. All the experiments in this study were implemented using Python 3.8.16 and PyTorch 2.0.1, running on a server with an Intel Xeon Gold 5222 CPU at 3.8 GHz, 160-GB RAM, and an NVIDIA GeForce RTX 2080Ti 11-GB GPU.

To demonstrate the superiority of the proposed method over popular algorithms, a comparison is made with several representative methods: CEM [15], ACE [19], CSCR [9], BLTSC [55], LCNNCD [56], HTDIRN [53], and TSTTD [50]. All the comparison methods are adjusted to their optimal configurations, and a fair comparison is conducted under the same experimental conditions. Specifically, for CSCR, the window sizes (w_{in}, w_{out}) are set to (11, 5), and the regularization parameters (λ_1, λ_2) are chosen as (0.1, 1.0). The CSCR algorithm is implemented using MATLAB 2017b. For BLTSC, the threshold ϵ for selecting background samples is set to 0.15, with 75% of the samples randomly selected for training. The learning rate is set to 1×10^{-4}, the batch size is 50, and the total number of epochs is 500. Target suppression loss is calculated by maximizing the four smallest SAM values. In nonlinear background suppression, the parameter λ is set to 10 for all the datasets. The BLTSC algorithm is implemented in MATLAB 2017b for coarse detection and background selection, and Python 3.6.15 with TensorFlow 1.14 is used for background reconstruction via AE. For LCNNCD, the local density threshold is set to 50, the proportion of background removed from the target is set to 0.2, and the Pearson correlation coefficient is used to select three target samples. The learning rate during training is 1×10^{-4}, the batch size is 64, and the total number of epochs is 50. For HTD-IRN, the background subspace dimension m and sparse weight η_1 are set as $(m, \eta_1) = (30, 10)$, (30, 10), (10, 0.1), and (10, 0.1) for San Diego A, San Diego B, abu-urban-2, and abu-urban-4, respectively. The learning rate during model training is 1×10^{-3}, and the number of epochs is 5000. For TSTTD, the group length is set to 20, the depth of the transformer encoder is 4, and the embedding dimension is 64. The learning rate during training is 1×10^{-4}, with a total of 20 training epochs and a weight decay of 0. The optimal model parameters are selected during detection. The experimental environment for LCNNCD, HTD-IRN, and TSTTD is consistent with that of the proposed method, which minimizes the influence of environmental factors on the results.

D. Parameter Analysis

The proposed method primarily involves the following important hyperparameters: margin_1, which adjusts the minimum similarity margin between positive and negative samples; margin_2, which adjusts the minimum similarity lower bound between the prior sample and positive sample; β, which regulates the weight of the target intraclass loss in the total loss; and ε, which adjusts the weight of the binary cross-entropy loss in the total loss. To discuss the combined effect of these parameters on the algorithm, $\text{AUC}_{(P_F,\ P_D)}$, and AUC_{SNPR} are used as performance metrics. The 3-D bar chart in Fig. 5 illustrates that the higher the bar under the current parameter settings, the better the detection performance. Specifically, the values for margin_1 and margin_2 are set to 0.95, 0.96, 0.97, 0.98, 0.99, and 1.0. The β takes values of 0.01, 0.02, 0.03, 0.04, and 0.05, while ε is set to 0.1, 0.5, 1.0, 5.0, and 10.0. It is evident that when both margin_1 and margin_2 are 0.98, the method achieves superior performance on all the datasets. Similarly, when β and ε are set to 0.03 and 5, respectively, the method also shows improved performance. This indicates that the parameter selection data-independent. In addition, it can be seen that a larger value of β leads to better performance for the proposed method.

Fig. 6 illustrates the impact of different ω settings on the experimental results. Based on the experimental results, we set ω to 0.4, as it yields a U-shaped Beta distribution that maintains semantic consistency while introducing sufficient diversity to the mixed hyperspectral samples.

E. Results and Analysis

Fig. 4 presents the detection maps obtained by seven comparative methods across the four benchmark datasets. Compared with other methods, the proposed method demonstrates significant advantages in background suppression and target extraction. In terms of background suppression, ACE, BLTSC, TSTTD, and the proposed method all effectively suppress the background to a low level. However, the first three methods also suppress the target component, leading to lower target detection. The CEM algorithm does not fully suppress the background, resulting in a detection map that resembles a noisy image. CSCR and LCNNCD exhibit a large number of false positives and fail to adequately separate the target from the background. HTD-IRN can suppress the background in the San Diego A, San Diego B, and abu-urban-2 scenarios, but it does not effectively suppress the background in the abu-urban-4 dataset. This could be due to the HTD-IRN method lacking effective strategies for suppressing background in scenarios with a large number of targets. In terms of target saliency, LCNNCD, HTD-IRN, TSTTD, and the proposed method all effectively highlight target pixels. Although LCNNCD can detect target pixels, it fails to increase the distinction between target and background, which could result in both high $\text{AUC}_{(\tau,\ P_D)}$ and $\text{AUC}_{(\tau,\ P_F)}$ values. HTD-IRN and TSTTD detect the targets, but neither extracts all the target pixels, resulting in incomplete shapes of targets in the detection maps. CEM, ACE, CSCR, and BLTSC fail to significantly highlight target pixels or reveal the target's shape. From the detection maps, it is evident that the proposed method can effectively extract target pixels and maximize the distinction between target and background pixels. The key reason for this is the augmentation of target pixel samples through the diffusion model, which enables the feature extractor to better recognize the differences and similarities between target and background.

The 3-D ROC curves and the three 2-D ROC curves for the four datasets are shown in Fig. 7. It can be observed that for all the four datasets, the 2-D ROC curves of (P_F, P_D) of the proposed method are located at the top-left corner of the plots, demonstrating the superior performance of the algorithm. Upon examining the 2-D ROC curves of (τ, P_D) across different datasets, it is evident that the proposed method consistently exhibits excellent performance. In addition, for most threshold settings, a stable and satisfactory (P_D) value

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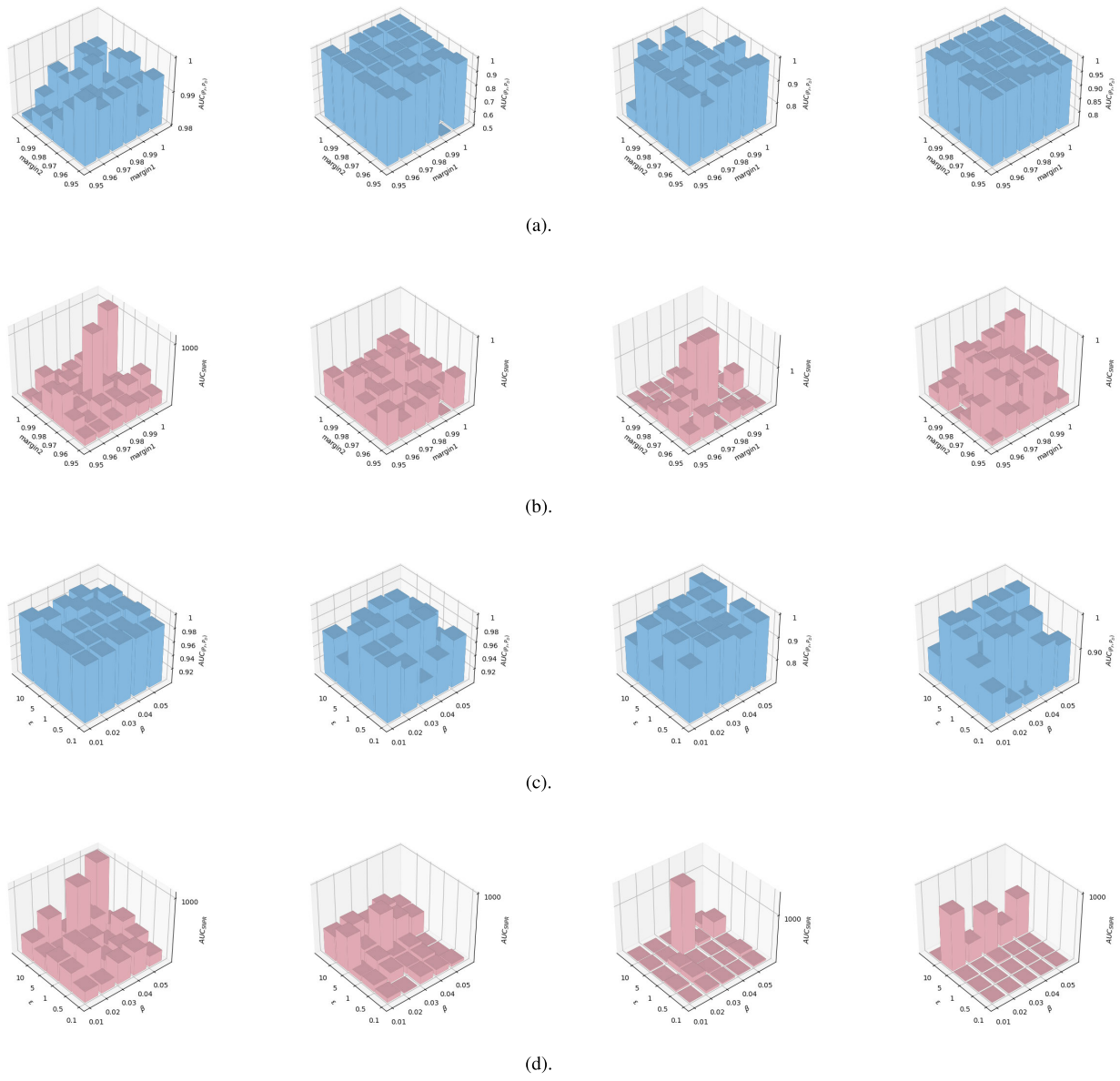


Fig. 5. Results on four datasets under different parameter settings. From left to right: San Diego A, San Diego B, abu-urban-2, and abu-urban-4.

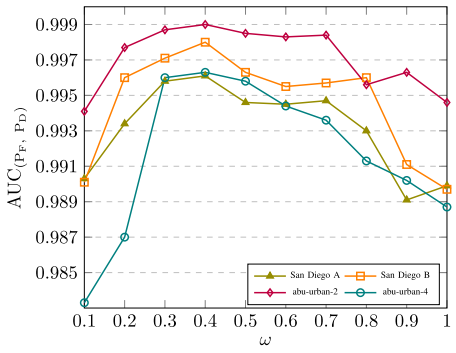


Fig. 6. Results of $\mathrm{AUC}_{(\mathrm{P_F},\ \mathrm{P_D})}$ under different ω settings.

is achieved, indicating that the algorithm is highly tolerant to threshold settings. In the 2-D ROC curves of (τ, P_F), the proposed method's curve is positioned in the bottom-left corner of the plot, significantly outperforming all the comparative algorithms, confirming the superior background suppression capability of the proposed method. Overall, the proposed method is more reliable compared with other methods.

Table I presents the quantitative target detection performance with the optimal results highlighted in bold and the second-best results underlined. Compared with other algorithms, the proposed method demonstrates accurate and robust performance across most quantitative metrics. For the $\mathrm{AUC}_{(\mathrm{P_F},\ \mathrm{P_D})}$, it is evident that the proposed method achieves the best results on all the four datasets. Notably, the LCNNCD algorithm achieves the best $\mathrm{AUC}_{(\tau,\ \mathrm{P_D})}$ values across all the four scenarios, and its $\mathrm{AUC}_{(\tau,\ \mathrm{P_F})}$ values also remain relatively high. However, the proximity of negative and positive sample distributions leads to a lower $\mathrm{AUC}_{(\mathrm{P_F},\ \mathrm{P_D})}$ score, which negatively impacts the overall performance of the LCNNCD algorithm. Considering all the quantitative metrics, the proposed algorithm excels at effectively

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TABLE I

QUANTITATIVE RESULTS OF THE COMPARATIVE METHODS ON THE FOUR DATASETS. BOLD INDICATES THE OPTIMAL RESULTS, AND UNDERLINE INDICATES THE SECOND-BEST RESULTS

HSI Data Set		CEM	ACE	CSCR	BLTSC	LCNNCD	HTD-IRN	TSTTD	HTD-DMCG
San Diego A	$AUC_{(P_F, P_D)}$	0.99611	0.98955	0.97659	0.99003	0.98351	0.98821	0.97099	**0.99612**
	$AUC_{(\tau, P_D)}$	0.45362	0.11502	0.55857	0.51209	**0.97740**	0.32967	0.61772	0.86152
	$AUC_{(\tau, P_F)}$	0.18647	0.00409	0.34982	0.07647	0.75440	0.00828	0.00117	**0.00080**
	AUC_{OA}	1.26326	1.10048	1.18534	1.42566	1.20651	1.30960	1.58754	**1.85684**
	AUC_{SNPR}	2.43264	28.11562	1.59673	6.69689	1.29559	39.81522	527.96581	**1072.87920**
San Diego B	$AUC_{(P_F, P_D)}$	0.84702	0.75851	0.99317	0.82724	0.95847	0.99498	0.99273	**0.99800**
	$AUC_{(\tau, P_D)}$	0.31796	0.03432	0.64576	0.08991	**0.81508**	0.39329	0.16724	0.65996
	$AUC_{(\tau, P_F)}$	0.19030	0.00448	0.29306	0.00182	0.54158	0.00451	0.00116	**0.00110**
	AUC_{OA}	0.97468	0.78835	1.34587	0.91533	1.23197	1.38376	1.15881	**1.65687**
	AUC_{SNPR}	1.67085	7.65730	2.20352	49.32200	1.50501	87.20399	144.17241	**602.70685**
abu-urban-2	$AUC_{(P_F, P_D)}$	0.56209	0.65955	0.95345	0.69095	0.92469	0.96077	0.99434	**0.99901**
	$AUC_{(\tau, P_D)}$	0.25718	0.02724	0.66562	0.01137	**0.88638**	0.64531	0.34773	0.75197
	$AUC_{(\tau, P_F)}$	0.21704	0.00283	0.27898	**0.00040**	0.55378	0.10791	0.00339	0.00052
	AUC_{OA}	0.60223	0.68395	1.34008	0.70191	1.25728	1.49817	1.33868	**1.75047**
	AUC_{SNPR}	1.18498	9.62063	2.38589	28.13861	1.60059	5.98008	102.57522	**1457.30814**
abu-urban-4	$AUC_{(P_F, P_D)}$	0.61509	0.63470	0.97206	0.78434	0.91972	0.94920	0.99519	**0.99633**
	$AUC_{(\tau, P_D)}$	0.23242	0.02622	0.66766	0.03341	**0.94097**	0.79711	0.32247	0.65131
	$AUC_{(\tau, P_F)}$	0.19189	0.00310	0.48757	**0.00034**	0.78858	0.29162	0.00402	0.00108
	AUC_{OA}	0.65562	0.65782	1.15215	0.81740	1.07211	1.45469	1.31364	**1.64656**
	AUC_{SNPR}	1.21118	8.45950	1.36936	97.11919	1.19325	2.73339	80.21642	**601.94640**

separating targets from the background while minimizing false alarms.

Fig. 8 presents the boxplots of the proposed method and the comparative methods. It is evident that the proposed method achieves a significant distance between target and background, indicating the superior background suppression capability of the method. The results from the boxplot are also validated by the ROC curves and the detection maps. Taking the San Diego A dataset as an example, the boxplot shows no overlap between the target and background boxes for the proposed method, which corresponds to the curve closely positioned at the top-right corner of the (τ, PD) 2-D ROC curve and the curve positioned at the bottom-left corner of the (τ, PF) 2-D ROC curve.

F. Ablation Study

This section validates each individual component of the proposed method. To ensure fairness, all other structures and experimental settings remain consistent, except for the discussed module.

1) Analysis of the Effectiveness of Data Augmentation: First, an ablation analysis is conducted on the coarse detection method used in the data augmentation process. Table II presents the results under different detection methods. Specifically, in addition to the traditional method CEM used in the proposed approach, the other two conventional algorithms, ACE and SAM, are used for comparison, denoted as "CEM-base," "ACE-base," and "SAM-base." It is evident that using the CEM method for coarse detection significantly reduces the $AUC_{(\tau, P_F)}$ values compared with the other two methods, thus improving the detection accuracy.

TABLE II

QUANTITATIVE RESULTS UNDER DIFFERENT COARSE DETECTION METHODS. BOLD INDICATES THE OPTIMAL RESULTS, AND UNDERLINE INDICATES THE SECOND-BEST RESULTS

HSI Data Set		ACE-base	SAM-base	CEM-base
San Diego A	$AUC_{(P_F, P_D)}$	0.98796	0.97386	**0.99612**
	$AUC_{(\tau, P_D)}$	0.94737	**0.98240**	0.86152
	$AUC_{(\tau, P_F)}$	0.00231	0.05832	**0.00080**
	AUC_{OA}	**1.93302**	1.89793	1.85684
	AUC_{SNPR}	409.58712	16.84379	**1072.87920**
San Diego B	$AUC_{(P_F, P_D)}$	0.92775	0.61161	**0.99800**
	$AUC_{(\tau, P_D)}$	**0.99999**	0.96436	0.65996
	$AUC_{(\tau, P_F)}$	0.89428	0.98902	**0.00110**
	AUC_{OA}	1.03346	0.58695	**1.65687**
	AUC_{SNPR}	1.1821	0.97507	**602.70685**
abu-urban-2	$AUC_{(P_F, P_D)}$	0.93732	0.80720	**0.99901**
	$AUC_{(\tau, P_D)}$	**0.94981**	0.93336	0.75197
	$AUC_{(\tau, P_F)}$	0.77311	0.99687	**0.00052**
	AUC_{OA}	1.11402	0.74368	**1.75047**
	AUC_{SNPR}	1.22855	0.93629	**1457.30814**
abu-urban-4	$AUC_{(P_F, P_D)}$	0.99325	0.69748	**0.99633**
	$AUC_{(\tau, P_D)}$	**0.82450**	0.81686	0.65131
	$AUC_{(\tau, P_F)}$	0.00324	0.93827	**0.00108**
	AUC_{OA}	**1.81451**	0.57608	1.64656
	AUC_{SNPR}	254.16215	0.87060	**601.94640**

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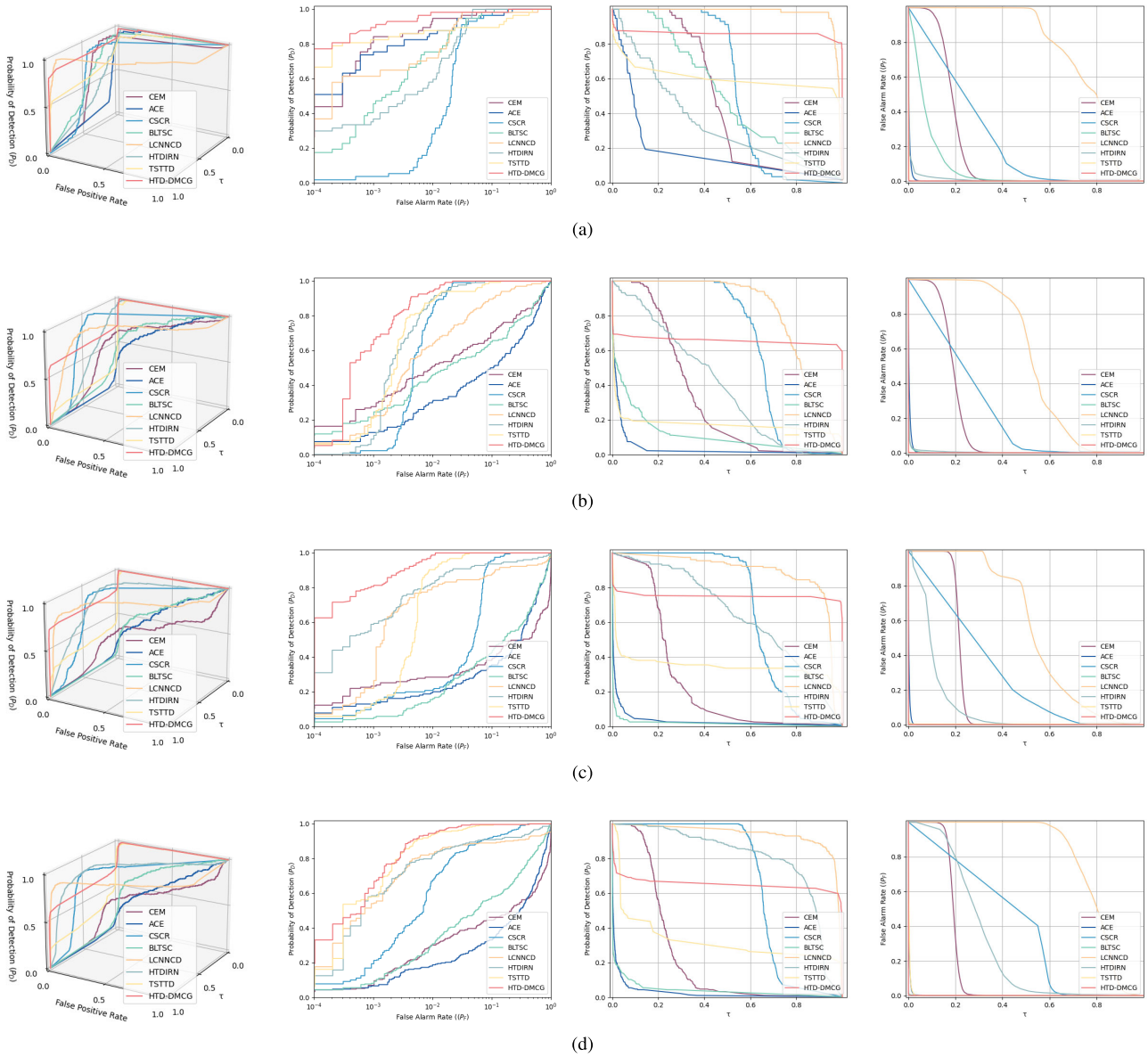


Fig. 7. ROC curves of the comparative methods on (a) San Diego A, (b) San Diego B, (c) abu-urban-2, and (d) abu-urban-4. (Left to right) 3-D ROC curve, 2-D ROC curve of (P_F, P_D), 2-D ROC curve of (τ, P_F), and 2-D ROC curve of (τ, P_D).

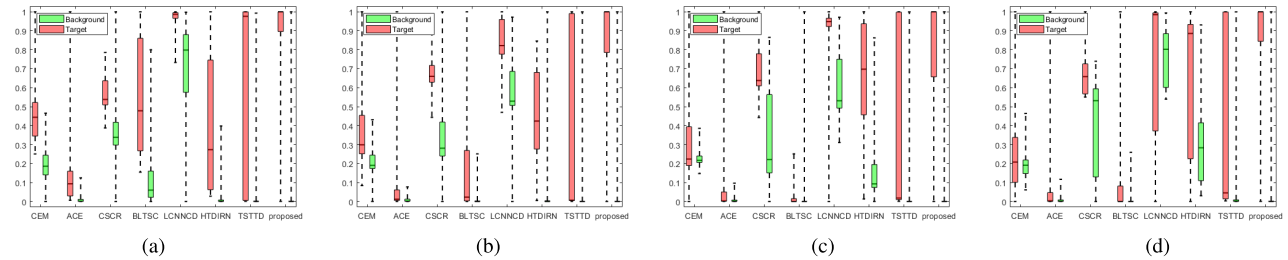


Fig. 8. Background–target separation diagrams of competing methods on (a) San Diego A, (b) San Diego B, (c) abu-urban-2, and (d) abu-urban-4.

Fig. 9 presents the direct detection results using CEM, ACE, and SAM methods as coarse detectors across four datasets. These three methods are referred to as "CEM-base," "ACE-base," and "SAM-base," respectively. Taking the San Diego A dataset as an example, when the SAM method is used as the coarse detection method, the $AUC_{(\tau,\ P_F)}$ values are 0.05832, as shown in Table II, and the detection results are suboptimal, with a significant number of false positives. In addition, using ACE and SAM methods for coarse detection leads to misidentification of background as targets. Therefore, the choice of coarse detection method is highly correlated with the quality of samples generated by the diffusion model. Specifically,

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TABLE III
QUANTITATIVE RESULTS UNDER DIFFERENT MODULE METHODS. BOLD INDICATES THE OPTIMAL RESULTS, AND UNDERLINE INDICATES THE SECOND-BEST RESULTS

	Module			San Diego A		San Diego B		abu-urban-2		abu-urban-4	
	DA	CGTE	TABS	$AUC_{(P_F,\ P_D)}$	$AUC_{(\tau,\ P_F)}$	$AUC_{(P_F,\ P_D)}$	$AUC_{(\tau,\ P_F)}$	$AUC_{(P_F,\ P_D)}$	$AUC_{(\tau,\ P_F)}$	$AUC_{(P_F,\ P_D)}$	$AUC_{(\tau,\ P_F)}$
(1)				0.73438	0.81218	0.64971	0.06867	0.99894	0.01291	0.91947	0.94547
(2)	✓			0.90743	0.03713	0.99558	0.01322	0.99879	0.01570	0.78772	0.47208
(3)	✓	✓		0.98426	0.00770	0.99623	0.01316	**0.99916**	0.00110	0.91303	0.40553
(4)	✓	✓	✓	**0.99612**	**0.00080**	**0.99800**	**0.00110**	0.99901	**0.00052**	**0.99633**	**0.00108**

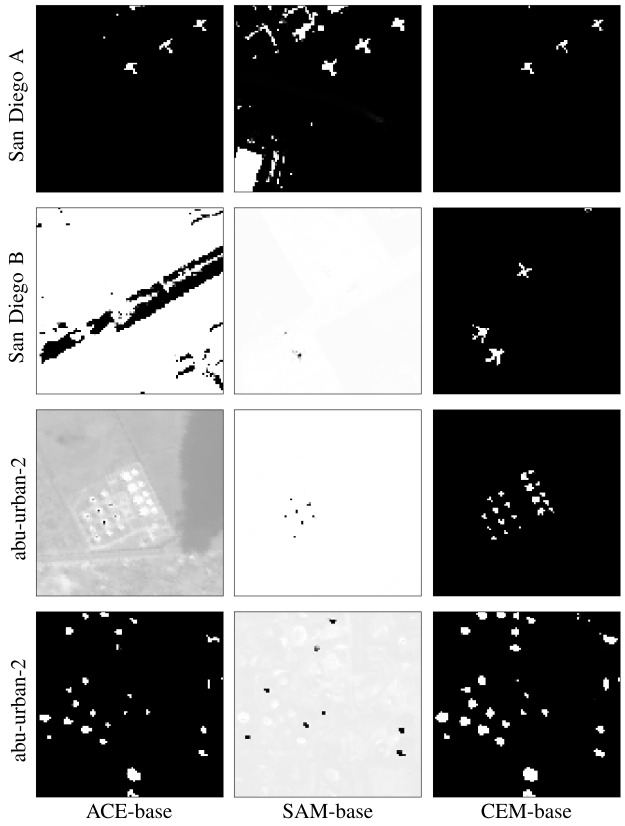


Fig. 9. Visualized results of different coarse detection methods on four datasets.

a more stable and effective coarse detection method increases the likelihood of selecting true target pixels as pseudotargets. Consequently, fewer noisy samples are included in the training data for the denoising model, enabling the model to more effectively learn spectral features of the target class.

2) Analysis of the Effectiveness of Different Modules: The key contributions of the proposed method lie in the data augmentation technique, the use of convolutional GLU to replace the feedforward network in the transformer, and the TABS loss function. To better analyze the impact of these methods, we evaluated their contributions through ablation experiments. As shown in Table III, we investigate four scenarios: 1) no data augmentation (DA) using diffusion models and MixUp, no gated convolution GLU transformer encoder (CGTE), and no TABS loss (TABS); 2) using data augmentation only; 3) using convolutional GLU to replace the feedforward network in the transformer on top of 2); and 4) adding TABS loss on top of 3). Scenario 4) significantly improves the model's quantitative metrics, $AUC_{(P_F,\ P_D)}$ enabling more accurate discrimination between target and background. This indicates that convolutional GLU can capture information between adjacent spectral bands, resulting in more discriminative spectral features. Scenario 4) also notably reduces $AUC_{(\tau,\ P_F)}$ compared with 3), suggesting that the target intraclass loss brings the target samples closer together in the feature space as desired.

IV. CONCLUSION

This article presents a method designed for HTD, named HTD-DMCG. First, data augmentation is performed using diffusion models and MixUp techniques to generate sufficient and balanced training samples. Then, a convolutional GLU-based feature extraction network is used, enabling the model to learn more critical information from adjacent spectral bands for representation. Furthermore, a novel TABS loss is proposed to enhance the separability between target and background, as well as the aggregation within the target class. Ablation experiments demonstrate that the TABS loss improves the aggregation within the target class, thereby improving detection performance. The experimental results demonstrate that the HTD-DMCG outperforms state-of-the-art methods across all the four datasets.

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